

Experiment: Breadth First Search (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal on a graph.

Objective

To traverse all the nodes of a graph level by level using the BFS algorithm and understand the working of queue-based graph traversal.

Algorithm

- Create a graph using an adjacency list.
- Initialize a queue with the starting node.
- Mark the starting node as visited.
- Remove a node from the queue.
- Print the node.
- Add all unvisited adjacent nodes to the queue.
- Repeat steps 4–6 until the queue becomes empty.

Code

Python

Run

```
from collections import deque
```

```
g = {'A':['B','C'], 'B':['D'], 'C':['E'], 'D':[], 'E':[]}
```

```
q = deque(['A'])
```

```
v = set(['A'])
```

```
print("BFS Traversal:")
```

```
while q:
```

```
    n = q.popleft()
```

```
    print(n, end=' ')
```

```
    for i in g[n]:
```

```
        if i not in v:
```

```
v.add(i)  
q.append(i)
```

Output

BFS Traversal:

A B C D E

Result

The BFS traversal of the given graph was successfully performed using Python, and the nodes were visited in the order: **A → B → C → D → E**.

Conclusion

The Breadth First Search algorithm was implemented successfully in Python. BFS visits graph nodes level by level using a queue and is useful for shortest-path and graph traversal applications.

ChatGPT can make mistakes. Check important info.